

Brad Cai

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A maroon square containing the white letters "BC" in a serif font.

Education

Bachelor of Science in Computer Science UC Riverside - Bourns College of Engineering

Riverside, California — September 2015 to June 2019

- Graduated cum laude with a GPA of 3.63.
- Taken courses in Computer Graphics, GPU Programming, Virtual Reality, Graphics and Electronic Games, Linear Algebra, Concurrent Programming, C++, Intermediate Algorithms, Operating Systems, Data Structures, Software Design, Compiler Design, and Machine Learning.

Experience

Software Engineer, ServiceNow

San Diego, California — August 2019 to January 2025

- Worked with Java, Javascript, AngularJS, SQL, syntax parsing, and software engineering design principles to contribute to the architecture and implementation of various projects in ServiceNow's multi-node server product.
- Lead the development of the Instance Scan linter check feature and created software architecture plans for it that were approved by the internal architecture review board.
- Used Java Flight Recorder and the IntelliJ profiler to generate performance data to analyze & fix performance issues.
- Wrote tests, triaged tests and fixed defects for core platform stability within a large code base.

Research Assistant, UC Riverside

Riverside, California — October 2017 to December 2017

- Used iTensor, C++, and Python to create a tool for calculating quantum wave functions.

Skills

Computer Graphics

- Vertex, Fragment, and Compute Shader programming in HLSL and GLSL.
- DirectX 11 / D3D11 and Vulkan graphics APIs.
- Graphics pipeline and real-time rendering techniques.
- Unity engine and Godot engine experience.
- Vector math, Linear Algebra, and Trigonometry.

Software Engineering

- C++, C#, C, Java, Python, Javascript, and SQL.
- Software design principles and patterns.
- Performance / memory profiling and optimization.
- Multithreading and asynchronous programming.
- GNU/Linux environments and development tools.